A Hybrid Quantum-Classical Attention Mechanism for Efficient LLMs

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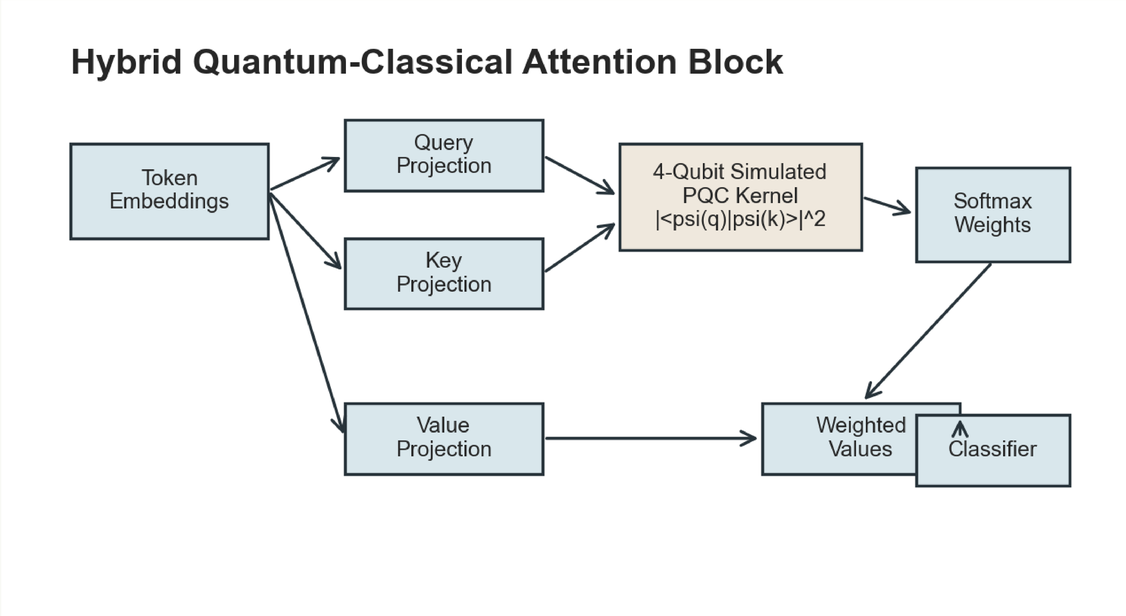
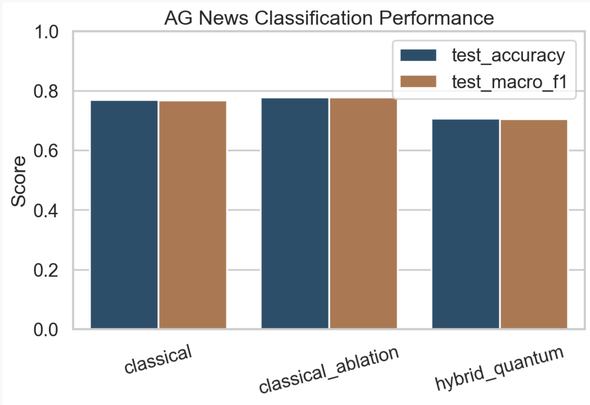
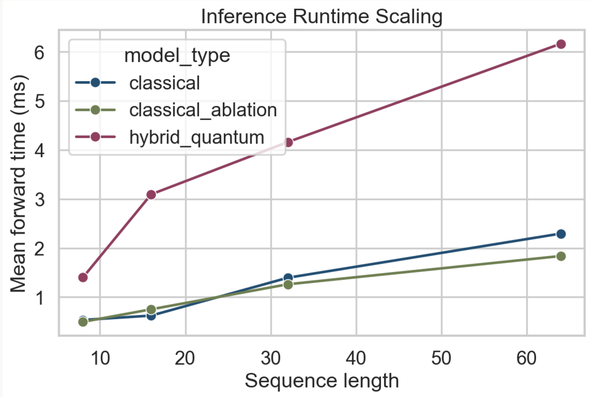
Research Question
Can a simulated quantum kernel replace only query-key similarity while preserving attention behavior?

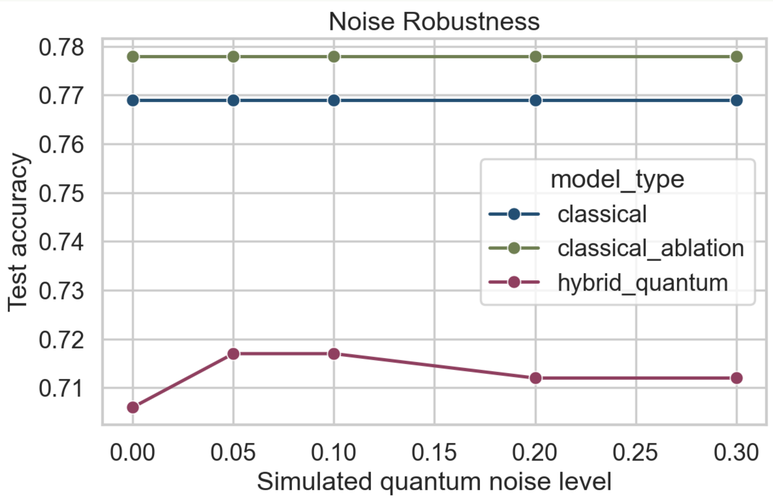
Main Result
Classical: 76.9%
Hybrid: 70.6%
Ablation: 77.8%
Hybrid runtime ratio: 2.99x

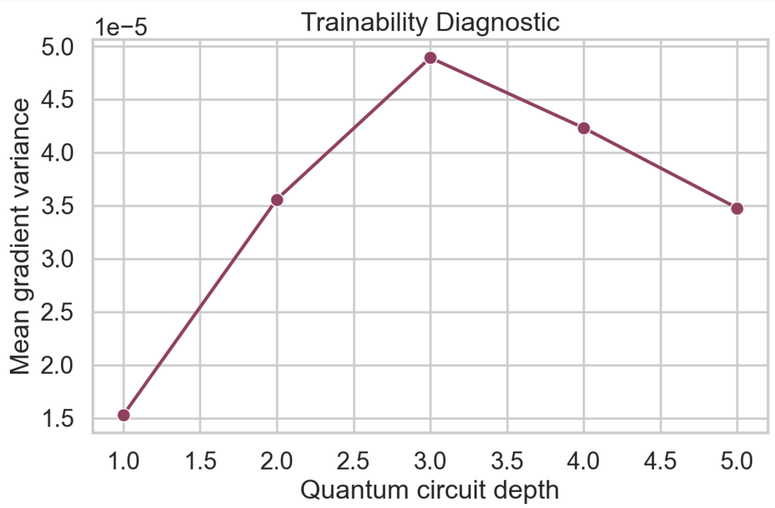
Conclusion
The hybrid layer is feasible, but state-vector simulation is slower. The framework is most useful for diagnostic comparison.

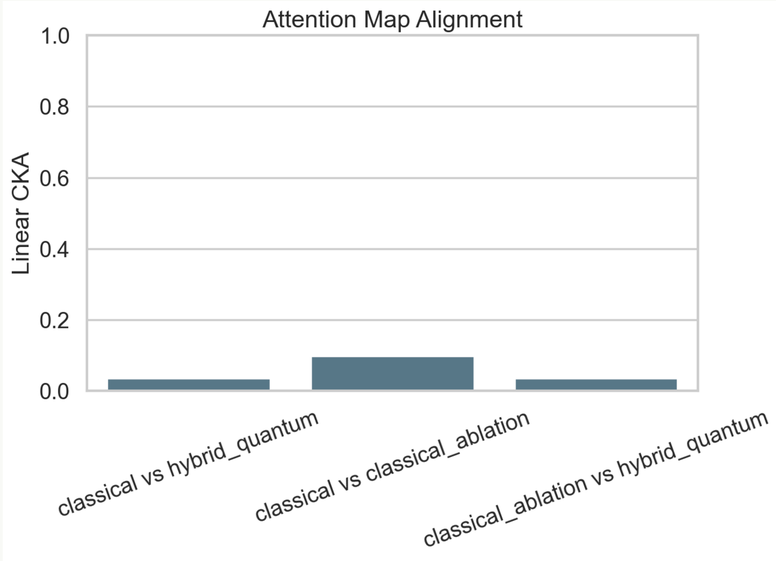












Metrics are generated by scripts/run_all.py and saved as CSV files in results/metrics/. Negative or mixed findings are reported directly.